

# Git and GitHub

Submitted to Vikul\_Sir

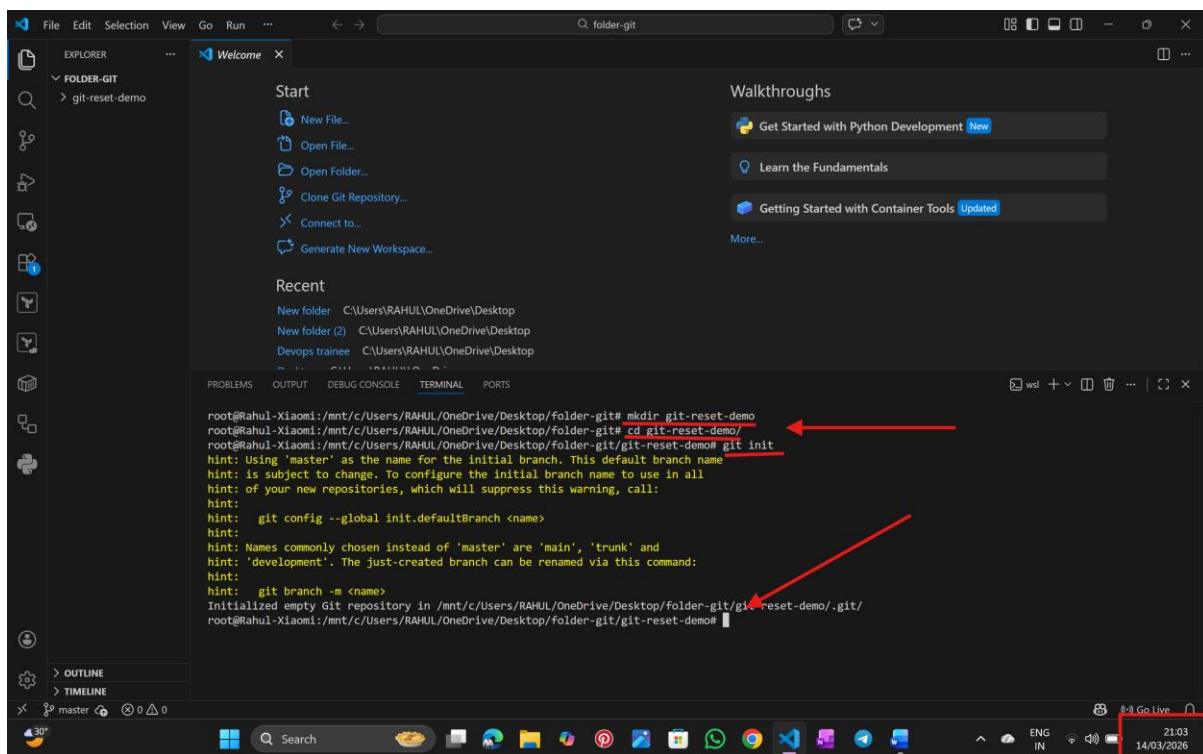
Submitted by Nalathati Rahul

Submission Date –16/MAR/2026

L1 - Create Local git repository and demonstrate all git reset options and revert. Compare the differences

Step :1 Create a Local Git Repository

- open visual studio code
- make a directory use command -- mkdir git-reset-demo
- change to the directory use command -- cd git-reset-demo
- initialization of git use command -- git init



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the following commands and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# mkdir git-reset-demo
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# cd git-reset-demo/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo/.git/
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

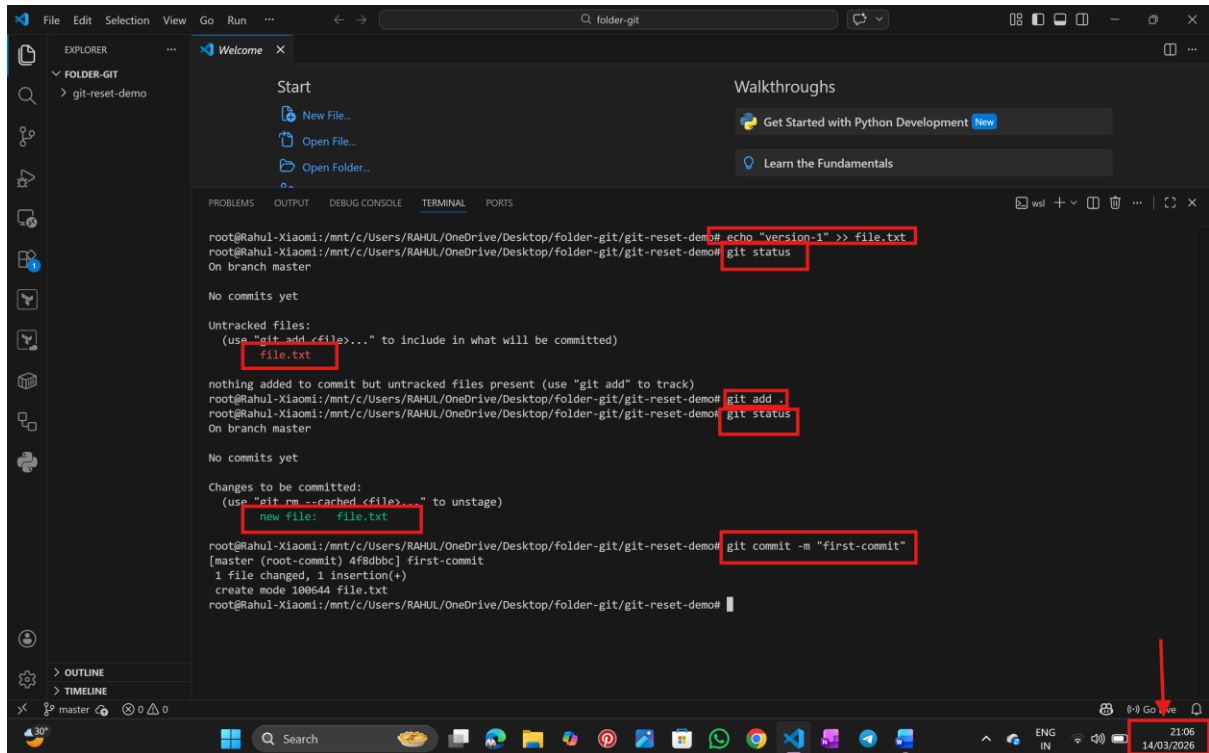
Red arrows point to the `mkdir git-reset-demo`, `cd git-reset-demo/`, and `git init` commands in the terminal. The system clock in the bottom right corner shows 21:03 on 14/03/2026.

Step :2 Now create some files and make some commits inside the directory

echo "Version -1" > file.txt

git add <file.txt> or can use <.>

git commit -m "First-commit"



The screenshot shows a VS Code window with a terminal open. The terminal output is as follows:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# echo "version-1" >> file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file.txt

nothing added to commit but untracked files present (use "git add" to track)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file.txt

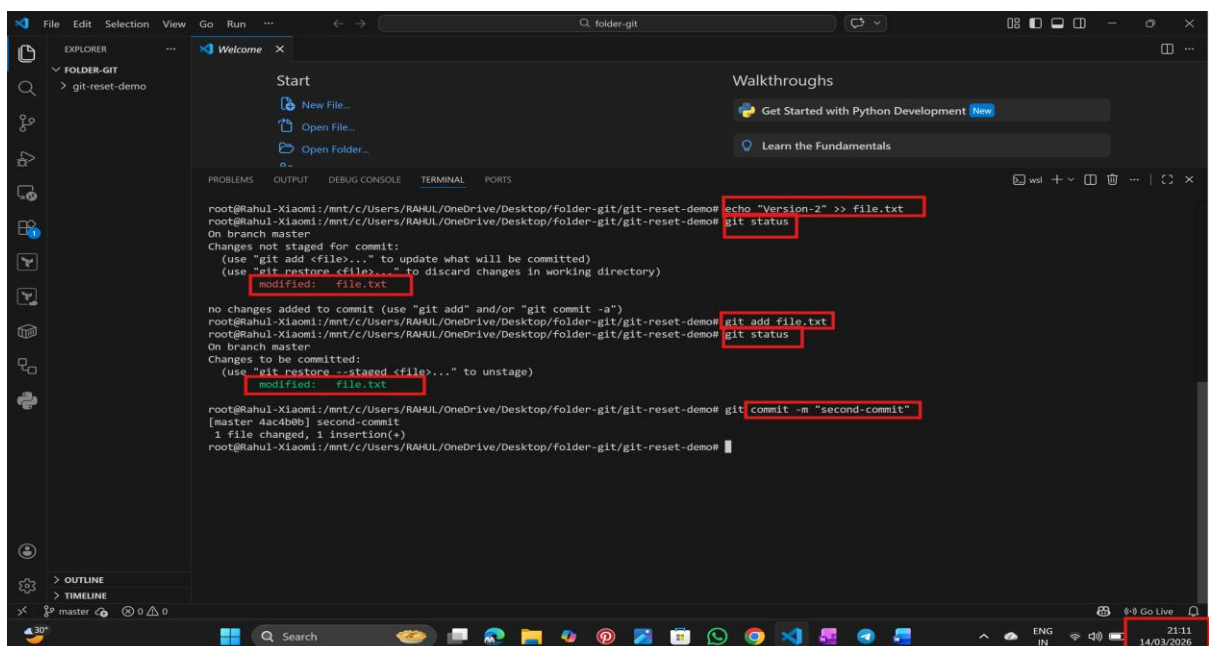
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git commit -m "first-commit"
[master (root-commit) 4f8dbbc] first-commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

The terminal window is titled "Welcome" and has tabs for "Start" and "Walkthroughs". The "Start" tab is active, showing the "New File..." option. The "Walkthroughs" tab is also visible, showing "Get Started with Python Development" and "Learn the Fundamentals". The terminal output is highlighted with red boxes. The system tray at the bottom right shows the time as 21:05 and the date as 14/03/2026.

echo "Version -2" >> file.txt

git add <file.txt> or can use <.>

git commit -m "Second commit"



The screenshot shows a VS Code window with a terminal open. The terminal output is as follows:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# echo "Version-2" >> file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   file.txt

no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git add file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   file.txt

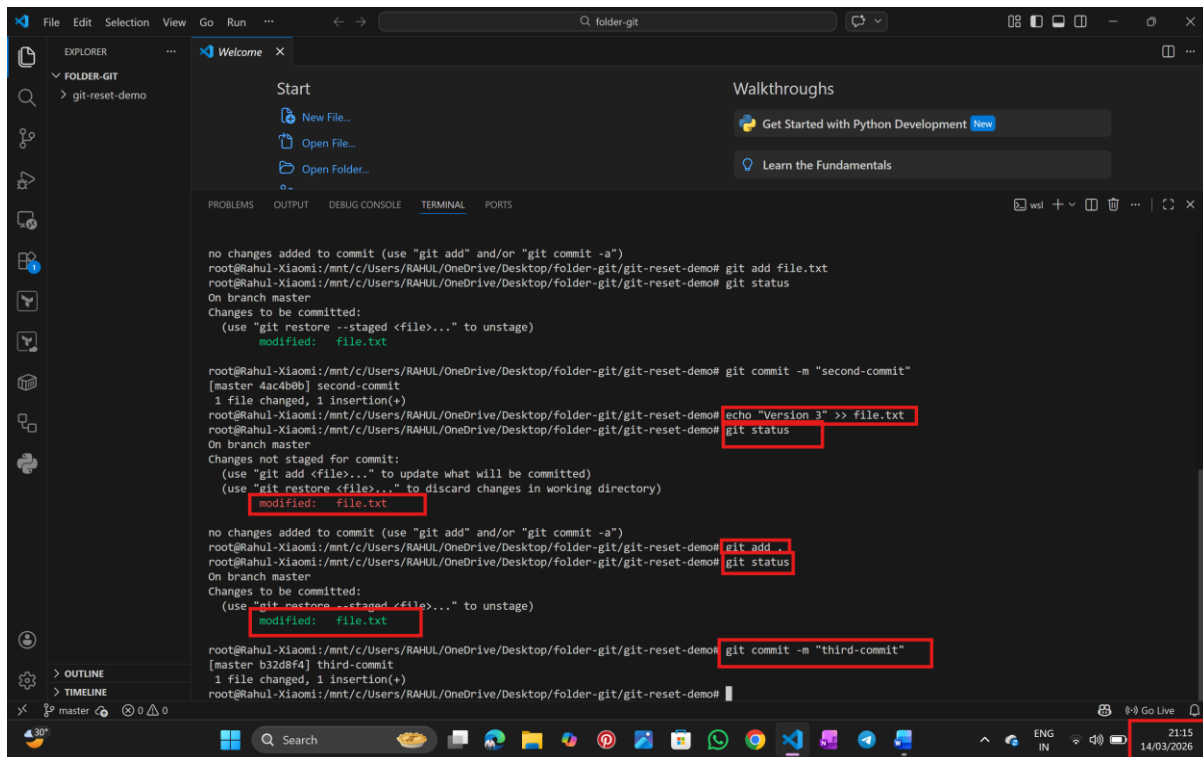
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git commit -m "second-commit"
[master 4ac400b] second-commit
1 file changed, 1 insertion(+)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

The terminal window is titled "Welcome" and has tabs for "Start" and "Walkthroughs". The "Start" tab is active, showing the "New File..." option. The "Walkthroughs" tab is also visible, showing "Get Started with Python Development" and "Learn the Fundamentals". The terminal output is highlighted with red boxes. The system tray at the bottom right shows the time as 21:11 and the date as 14/03/2026.

echo "Version 3" >> file.txt

git add <file.txt> or can use <.>

git commit -m "Third commit"



The screenshot shows a VS Code window with a terminal open. The terminal displays the following commands and output:

```
no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git add file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git commit -m "second-commit"
[master 4ac4b8b] second-commit
 1 file changed, 1 insertion(+)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# echo "Version 3" >> file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.txt

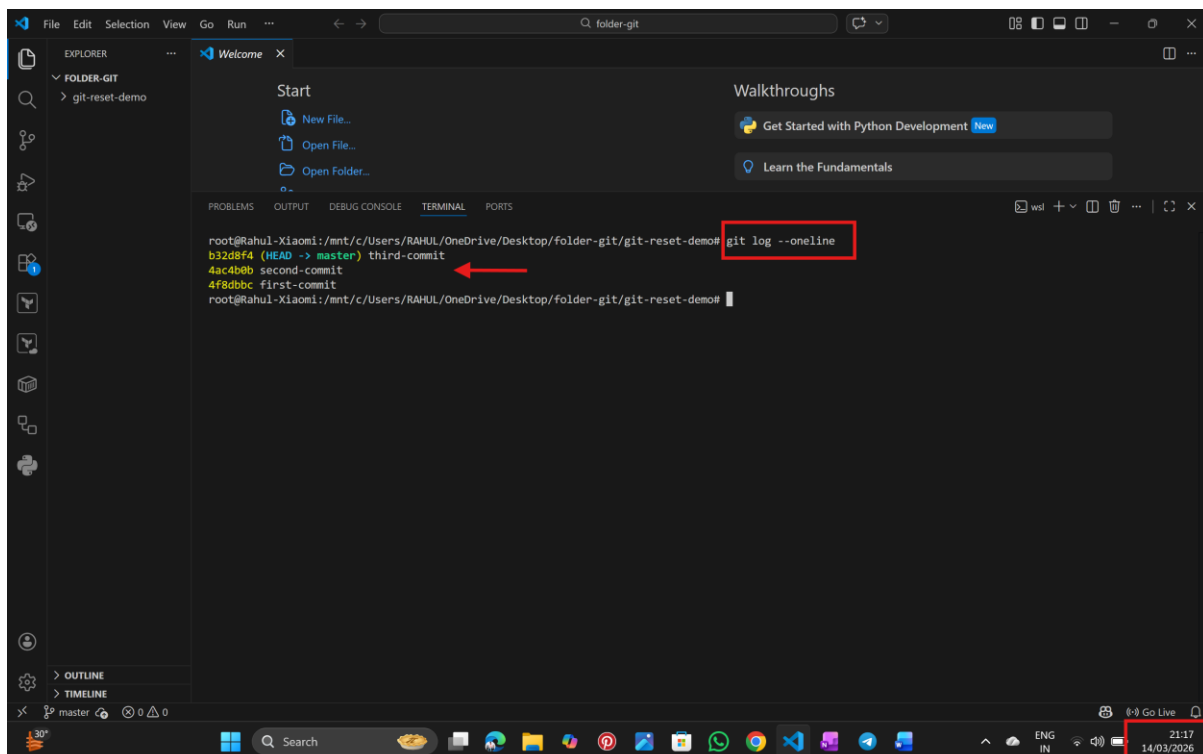
no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git add .
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt

root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git commit -m "third-commit"
[master b32dbf4] third-commit
 1 file changed, 1 insertion(+)
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

The status bar at the bottom right shows the time 21:15 and date 14/03/2026.

Step: 3 Now check the history of the commits

→ using command – git log ---oneline



The screenshot shows a VS Code window with a terminal open. The terminal displays the following command and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
b32dbf4 (HEAD -> master) third-commit
4ac4b8b second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

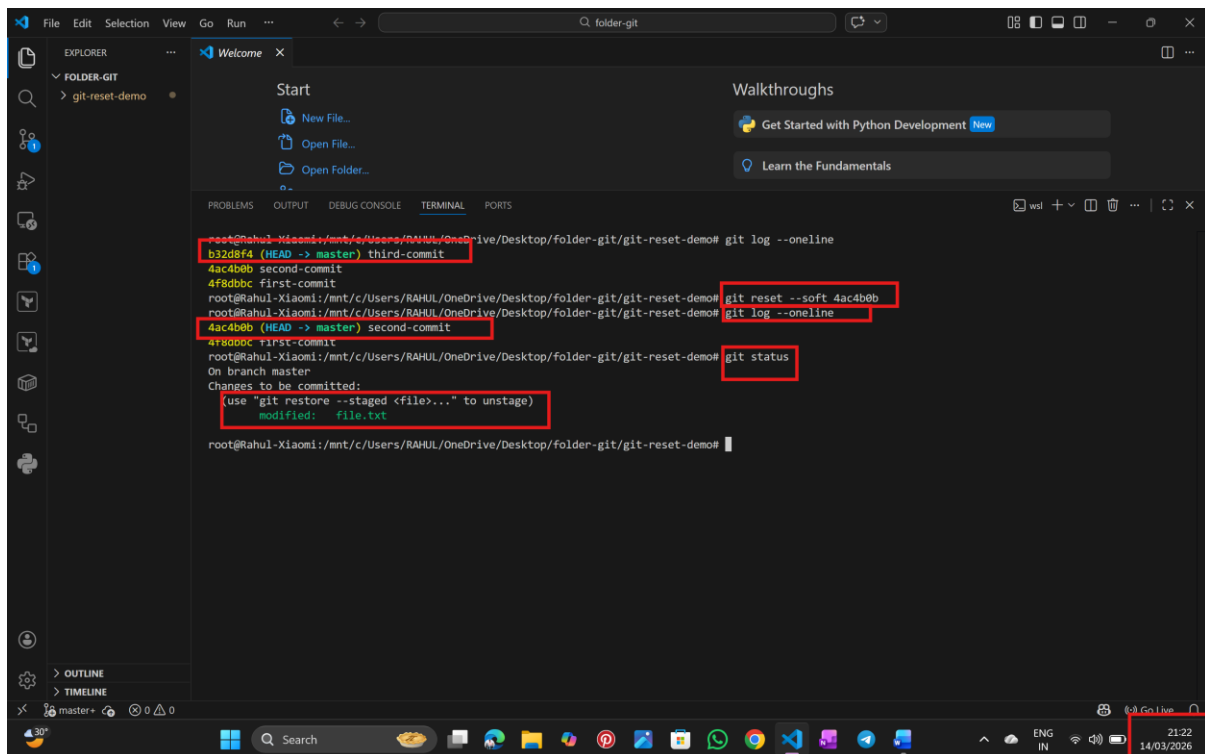
The status bar at the bottom right shows the time 21:17 and date 14/03/2026.

#### Step: 4 Now work on Git reset

→git reset used to move the HEAD pointer to previous commit. There are 3 types generally

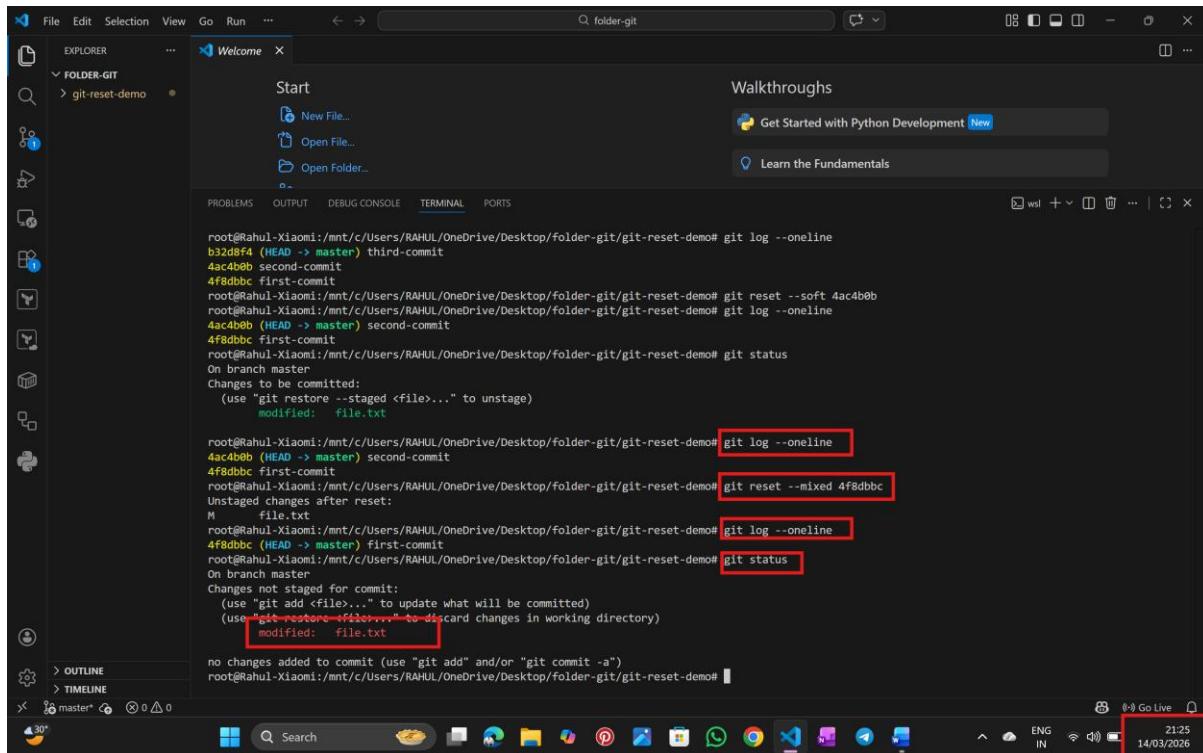
1. Git reset –soft
2. Git reset – mixed
3. Git reset – hard

→git reset –soft -- moves HEAD to a previous commit but keeps changes staged, Commit is removed, Files remain in staging area.



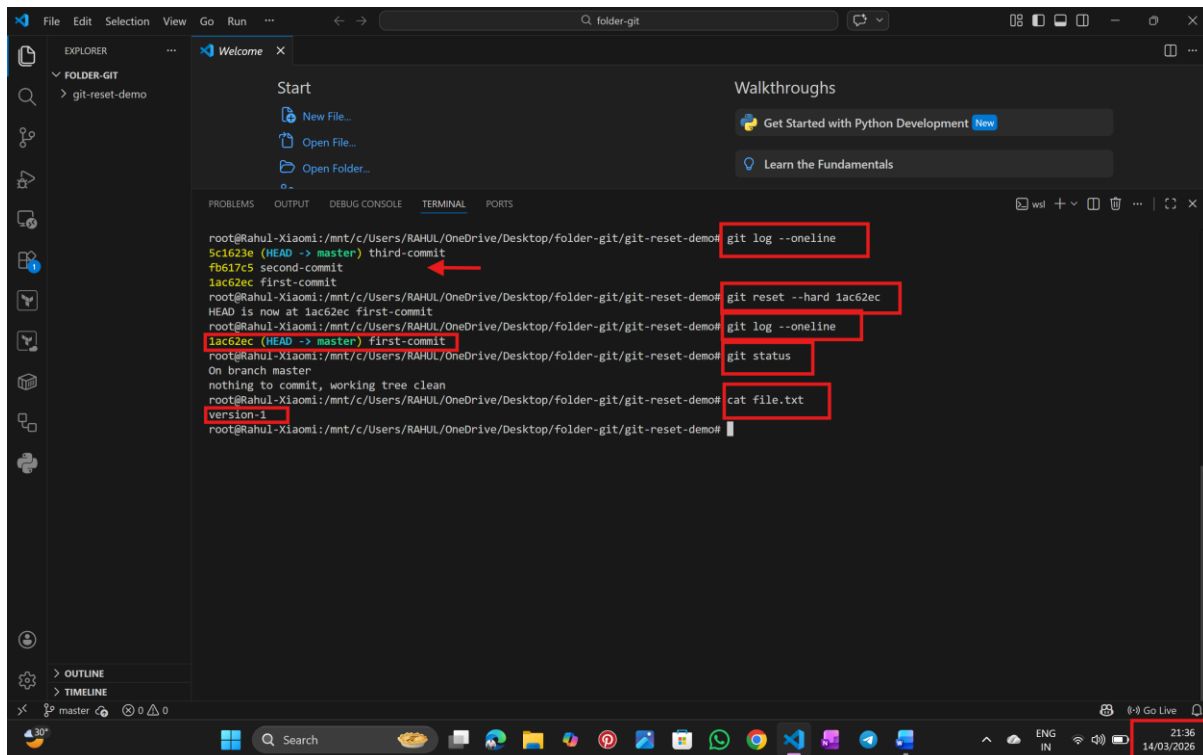
```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
b32d8f4 (HEAD -> master) third-commit
4ac4b0b second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git reset --soft 4ac4b0b
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
4ac4b0b (HEAD -> master) second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

→git reset --mixed moves HEAD and unstages the files but keeps the changes in the working directory.



```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
b32d8f4 (HEAD -> master) third-commit
4ac4b0b second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git reset --soft 4ac4b0b
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
4ac4b0b (HEAD -> master) second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
4ac4b0b (HEAD -> master) second-commit
4f8dbbc first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git reset --mixed 4f8dbbc
Unstaged changes after reset:
M   file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
4f8dbbc (HEAD -> master) first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore --staged <file>..." to discard changes in working directory)
        modified:   file.txt
no changes added to commit (use "git add" and/or "git commit -a")
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

→git reset --hard moves HEAD and deletes all changes permanently, commit removed, Changes deleted, working directory reset

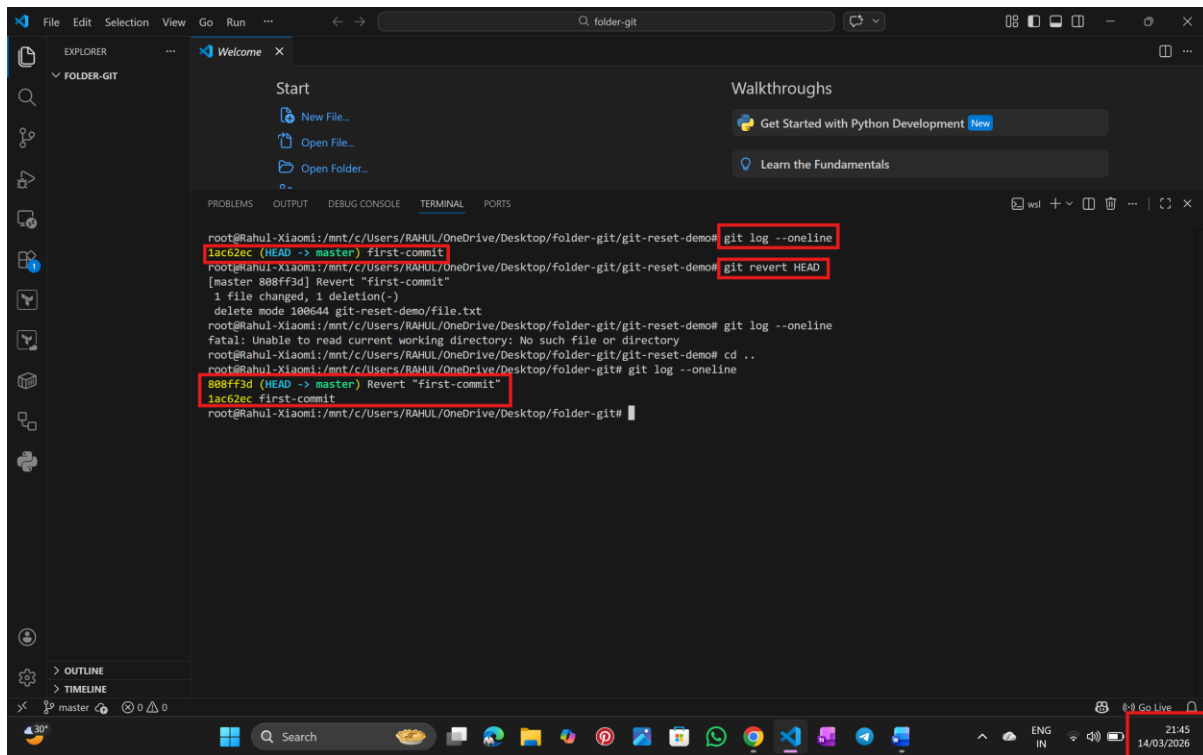


```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
5c1623e (HEAD -> master) third-commit
fb617c5 second-commit
1ac62ec first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git reset --hard 1ac62ec
HEAD is now at 1ac62ec first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
1ac62ec (HEAD -> master) first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git status
On branch master
nothing to commit, working tree clean
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# cat file.txt
version-1
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo#
```

Step: 5 Now work on git revert

→git revert is a Git command used to undo a specific commit by creating a new commit.

→ git revert is a Git command used to undo a specific commit by creating a new commit.



The screenshot shows a VS Code terminal window with the following commands and output:

```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
1ac62ec (HEAD -> master) first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git revert HEAD
[master 808ff3d] Revert "first-commit"
1 file changed, 1 deletion(-)
delete mode 100644 git-reset-demo/file.txt
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# git log --oneline
fatal: Unable to read current working directory: No such file or directory
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git/git-reset-demo# cd ..
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git# git log --oneline
808ff3d (HEAD -> master) Revert "first-commit"
1ac62ec first-commit
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/folder-git#
```

The terminal output shows that the first commit (1ac62ec) was successfully reverted, creating a new commit (808ff3d) that undoes the changes. The git log shows both commits, indicating that the history is preserved.

Step: 6 compare git reset and git revert

→git reset removes commits from history,

→whereas git revert keeps history and adds a new commit to undo changes.

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NALATHATI RAHUL